Transplantation & Oncology Critical Care – Chapter Summary

Donor Care • Recipients • Immunosuppression • Onco-Emergencies

ICU / EDIC Board Review • MCQ-Oriented

1. Organ Donation Pathways

- **DBD:** donation after brain death (neurological criteria)
- **DCD:** donation after circulatory death – withdrawal of life support, **no-touch period** (2–5 min, country-dependent) after asystole
- **Warm ischemia time** critical in DCD
- **NRP** (normothermic regional perfusion) improves organ quality
- Routine referral of all potential donors; separate donation discussion from withdrawal decision
- **Contraindications:** active untreated cancer (some exceptions), uncontrolled sepsis, prion disease

2. Physiology after Brain Death

- **Cushing reflex / catecholamine storm** → hypertension, tachycardia, myocardial injury
- Followed by **vasoplegia & hypotension**
- **Central DI** (80%) → hypernatremia, polyuria
- Hypothermia, coagulopathy (DIC), hyperglycemia
- Neurogenic pulmonary edema
- ↓ T3, cortisol

3. Donor Management Targets

Parameter	Target
MAP	60–110 (≥ 65)
CVP	4–10 mmHg
Urine output	0.5–3 mL/kg/h
Na	135–155
Glucose	4–10 mmol/L
SpO_2 / PaO_2	≥ 95% / > 80
Hb	> 7 g/dL
Temperature	> 35–36 °C

- **Vasopressin** first-line vasopressor (also treats DI); **desmopressin** for DI
- **Methylprednisolone 15 mg/kg** (lung donors)
- **Lung-protective ventilation** ↑ lungs procured; recruitment, bronchoscopy
- Thyroid hormone (T4): no benefit routinely

4. Immunosuppressants

Drug	Toxicity
Tacrolimus / ciclosporin (CNI)	AKI, HTN, ↑ K, ↓ Mg, PRES, tremor, TMA, diabetes
Mycophenolate	Marrow suppression, GI
Azathioprine	Myelosuppression (TPMT), pancreatitis
Sirolimus / everolimus	Wound healing, pneumonitis, ↑ lipids
Steroids	Hyperglycemia, infection, myopathy
Anti-thymocyte globulin	Cytokine release, cytopenias

- **CNI drug interactions:** azoles, macrolides ↑ levels; rifampicin, phenytoin ↓ levels

5. Infections Timeline after SOT

Period	Typical infections
< 1 month	Nosocomial (MDR bacteria, Candida, C. difficile), donor-derived
1–6 months	Opportunistic: CMV, PCP, Aspergillus, Nocardia, Listeria, Toxoplasma, BK virus, HBV/HCV reactivation
> 6 months	Community-acquired; late CMV, PTLD (EBV)

- **Prophylaxis:** TMP-SMX (PCP), valganciclovir (CMV), antifungals (lung / liver)
- **CMV:** fever, leukopenia, colitis, pneumonitis → ganciclovir

6. Organ-Specific Recipient Issues

- **Lung:** primary graft dysfunction (PGD 0–72 h, ARDS-like) → protective ventilation, iNO, ECMO; airway anastomosis dehiscence
- **Liver:** hepatic artery thrombosis (Doppler, ↑ LFT) → re-transplant; primary non-function
- **Kidney:** delayed graft function, renal artery / vein thrombosis, urine leak
- **Heart:** denervated (no atropine response, no angina) → isoprenaline / pacing; RV failure (↑ PVR)
- **Rejection:** hyperacute (minutes), acute cellular (weeks), antibody-mediated, chronic

7. GVHD & HSCT

- **Acute GVHD** (< 100 days classic): skin, liver (bilirubin), GI (diarrhea) → steroids, ruxolitinib
- **Engraftment syndrome:** fever, rash, capillary leak around neutrophil recovery → steroids
- **Sinusoidal obstruction syndrome (VOD):** hepatomegaly, jaundice, weight gain, ascites → defibrotide
- **Idiopathic pneumonia syndrome**, DAH, **TA-TMA**
- ICU outcome in HSCT improved; worse with ventilation + organ failure

8. Neutropenic Sepsis

- **Definition:** ANC < 0.5 × 10^9/L + fever ≥ 38.3 °C (or ≥ 38 × 1 h)
- **Antibiotics within 1 h:** anti-pseudomonal β-lactam (piperacillin-tazobactam / cefepime / meropenem)
- Add **vancomycin** if line infection, MRSA, skin infection, hemodynamic instability
- **Antifungal** if fever persists 4–7 days
- **G-CSF** not routine
- **MASCC score** ≥ 21 = low risk
- **Typhlitis (neutropenic enterocolitis):** RLQ pain, cecal wall thickening → conservative + broad antibiotics

9. Tumor Lysis Syndrome

- **Cairo-Bishop:** ↑ uric acid, ↑ K, ↑ PO_4, ↓ Ca (+ AKI, arrhythmia, seizure = clinical TLS)
- High risk: Burkitt, ALL, AML with high WBC, bulky lymphoma
- **Prevention:** hydration (UO > 100 mL/h), allopurinol (low–intermediate), **rasburicase** (high risk / uric acid ↑)
- **Rasburicase contraindicated in G6PD deficiency** (hemolysis, methemoglobinemia)
- No urine alkalinization
- Don't treat asymptomatic hypocalcemia ($CaPO_4$ precipitation)
- Early RRT if refractory

10. Oncologic Emergencies

- **Hyperleukocytosis / leukostasis** (WBC > 100k): hypoxemia, neuro → cytoreduction (hydroxyurea, leukapheresis); avoid transfusion
- **Superior vena cava syndrome:** stent, radiotherapy, steroids (lymphoma)
- **Malignant spinal cord compression:** dexamethasone 16 mg/day + MRI + radiotherapy / surgery
- **Hypercalcemia of malignancy:** saline, calcitonin, zoledronate, denosumab
- **SIADH:** small cell lung cancer
- **Malignant pericardial effusion:** tamponade → drainage
- **Airway obstruction:** rigid bronchoscopy, stent

11. Immunotherapy Toxicities

- **Cytokine release syndrome (CAR-T, bispecifics):** fever, hypotension, hypoxemia → **tocilizumab** ± steroids (ASTCT grading)
- **ICANS:** encephalopathy, seizures, cerebral edema → **steroids** (tocilizumab doesn't cross BBB)
- **Immune checkpoint inhibitors (irAEs):** pneumonitis, colitis, hepatitis, myocarditis, hypophysitis, adrenalitis → steroids (≥ 1–2 mg/kg), infliximab / mycophenolate if refractory
- **HLH:** ferritin ↑↑, cytopenias, fever → HScore; etoposide / dexamethasone, anakinra

12. Chemotherapy Toxicities

Drug	Toxicity
Anthracyclines (doxorubicin)	Cardiomyopathy
Bleomycin	Pulmonary fibrosis (avoid high FiO_2)
Cisplatin	AKI, ↓ Mg
Methotrexate	AKI, mucositis → glucarpidase, folinic acid
Ifosfamide	Encephalopathy (methylene blue), Fanconi
ATRA / arsenic	Differentiation syndrome → dexamethasone
5-FU / capecitabine	Coronary vasospasm (DPD deficiency)
Trastuzumab	Reversible cardiomyopathy

13. ICU Admission Decisions in Cancer

- Prognosis depends on **organ failures & performance status**, not cancer diagnosis alone
- **ICU trial:** full support 3–5 days then reassess
- Poor: > 3 organ failures after day 3, allo-HSCT with GVHD + MV
- Integrate **palliative care** early

14. High-Yield MCQ Pearls

- ☑ DCD requires no-touch period after asystole
- ☑ Brain-dead donor: vasopressin + desmopressin for DI
- ☑ Methylprednisolone 15 mg/kg in donors
- ☑ CNI toxicity: AKI, ↑K, PRES, TMA
- ☑ CMV peak 1–6 months post-transplant
- ☑ Denervated heart: atropine ineffective
- ☑ Neutropenic sepsis: anti-pseudomonal β-lactam ≤ 1 h
- ☑ Rasburicase contraindicated in G6PD deficiency
- ☑ CRS → tocilizumab; ICANS → steroids
- ☑ Leukostasis: avoid transfusion; leukapheresis
- ☑ ATRA → differentiation syndrome → dexamethasone
- ☑ Bleomycin → avoid high FiO_2

1. Organ Donation Pathways

- **DBD:** donation after brain death (neurological criteria)
- **DCD:** donation after circulatory death – withdrawal of life support, **no-touch period** (2–5 min, country-dependent) after asystole
- **Warm ischemia time** critical in DCD
- **NRP** (normothermic regional perfusion) improves organ quality
- Routine referral of all potential donors; separate donation discussion from withdrawal decision
- **Contraindications:** active untreated cancer (some exceptions), uncontrolled sepsis, prion disease

2. Physiology after Brain Death

- **Cushing reflex / catecholamine storm** → hypertension, tachycardia, myocardial injury
- Followed by **vasoplegia & hypotension**
- **Central DI** (80%) → hypernatremia, polyuria
- Hypothermia, coagulopathy (DIC), hyperglycemia
- Neurogenic pulmonary edema
- ↓ T3, cortisol

3. Donor Management Targets

Parameter	Target
MAP	60–110 (≥ 65)
CVP	4–10 mmHg
Urine output	0.5–3 mL/kg/h
Na	135–155
Glucose	4–10 mmol/L
SpO_2 / PaO_2	≥ 95% / > 80
Hb	> 7 g/dL
Temperature	> 35–36 °C

- **Vasopressin** first-line vasopressor (also treats DI); **desmopressin** for DI
- **Methylprednisolone 15 mg/kg** (lung donors)
- **Lung-protective ventilation** ↑ lungs procured; recruitment, bronchoscopy
- Thyroid hormone (T4): no benefit routinely

4. Immunosuppressants

Drug	Toxicity
Tacrolimus / ciclosporin (CNI)	AKI, HTN, ↑ K, ↓ Mg, PRES, tremor, TMA, diabetes
Mycophenolate	Marrow suppression, GI
Azathioprine	Myelosuppression (TPMT), pancreatitis
Sirolimus / everolimus	Wound healing, pneumonitis, ↑ lipids
Steroids	Hyperglycemia, infection, myopathy
Anti-thymocyte globulin	Cytokine release, cytopenias

- **CNI drug interactions:** azoles, macrolides ↑ levels; rifampicin, phenytoin ↓ levels

5. Infections Timeline after SOT

Period	Typical infections
< 1 month	Nosocomial (MDR bacteria, Candida, C. difficile), donor-derived
1–6 months	Opportunistic: CMV, PCP, Aspergillus, Nocardia, Listeria, Toxoplasma, BK virus, HBV/HCV reactivation
> 6 months	Community-acquired; late CMV, PTLD (EBV)

- **Prophylaxis:** TMP-SMX (PCP), valganciclovir (CMV), antifungals (lung / liver)
- **CMV:** fever, leukopenia, colitis, pneumonitis → ganciclovir

6. Organ-Specific Recipient Issues

- **Lung:** primary graft dysfunction (PGD 0–72 h, ARDS-like) → protective ventilation, iNO, ECMO; airway anastomosis dehiscence
- **Liver:** hepatic artery thrombosis (Doppler, ↑ LFT) → re-transplant; primary non-function
- **Kidney:** delayed graft function, renal artery / vein thrombosis, urine leak
- **Heart:** denervated (no atropine response, no angina) → isoprenaline / pacing; RV failure (↑ PVR)
- **Rejection:** hyperacute (minutes), acute cellular (weeks), antibody-mediated, chronic

7. GVHD & HSCT

- **Acute GVHD** (< 100 days classic): skin, liver (bilirubin), GI (diarrhea) → steroids, ruxolitinib
- **Engraftment syndrome:** fever, rash, capillary leak around neutrophil recovery → steroids
- **Sinusoidal obstruction syndrome (VOD):** hepatomegaly, jaundice, weight gain, ascites → defibrotide
- **Idiopathic pneumonia syndrome**, DAH, **TA-TMA**
- ICU outcome in HSCT improved; worse with ventilation + organ failure

8. Neutropenic Sepsis

- **Definition:** ANC < 0.5 × 10^9/L + fever ≥ 38.3 °C (or ≥ 38 × 1 h)
- **Antibiotics within 1 h:** anti-pseudomonal β-lactam (piperacillin-tazobactam / cefepime / meropenem)
- Add **vancomycin** if line infection, MRSA, skin infection, hemodynamic instability
- **Antifungal** if fever persists 4–7 days
- **G-CSF** not routine
- **MASCC score** ≥ 21 = low risk
- **Typhlitis (neutropenic enterocolitis):** RLQ pain, cecal wall thickening → conservative + broad antibiotics

9. Tumor Lysis Syndrome

- **Cairo-Bishop:** ↑ uric acid, ↑ K, ↑ PO_4, ↓ Ca (+ AKI, arrhythmia, seizure = clinical TLS)
- High risk: Burkitt, ALL, AML with high WBC, bulky lymphoma
- **Prevention:** hydration (UO > 100 mL/h), allopurinol (low–intermediate), **rasburicase** (high risk / uric acid ↑)
- **Rasburicase contraindicated in G6PD deficiency** (hemolysis, methemoglobinemia)
- No urine alkalinization
- Don't treat asymptomatic hypocalcemia ($CaPO_4$ precipitation)
- Early RRT if refractory

10. Oncologic Emergencies

- **Hyperleukocytosis / leukostasis** (WBC > 100k): hypoxemia, neuro → cytoreduction (hydroxyurea, leukapheresis); avoid transfusion
- **Superior vena cava syndrome:** stent, radiotherapy, steroids (lymphoma)
- **Malignant spinal cord compression:** dexamethasone 16 mg/day + MRI + radiotherapy / surgery
- **Hypercalcemia of malignancy:** saline, calcitonin, zoledronate, denosumab
- **SIADH:** small cell lung cancer
- **Malignant pericardial effusion:** tamponade → drainage
- **Airway obstruction:** rigid bronchoscopy, stent

11. Immunotherapy Toxicities

- **Cytokine release syndrome (CAR-T, bispecifics):** fever, hypotension, hypoxemia → **tocilizumab** ± steroids (ASTCT grading)
- **ICANS:** encephalopathy, seizures, cerebral edema → **steroids** (tocilizumab doesn't cross BBB)
- **Immune checkpoint inhibitors (irAEs):** pneumonitis, colitis, hepatitis, myocarditis, hypophysitis, adrenalitis → steroids (≥ 1–2 mg/kg), infliximab / mycophenolate if refractory
- **HLH:** ferritin ↑↑, cytopenias, fever → HScore; etoposide / dexamethasone, anakinra

12. Chemotherapy Toxicities

Drug	Toxicity
Anthracyclines (doxorubicin)	Cardiomyopathy
Bleomycin	Pulmonary fibrosis (avoid high FiO_2)
Cisplatin	AKI, ↓ Mg
Methotrexate	AKI, mucositis → glucarpidase, folinic acid
Ifosfamide	Encephalopathy (methylene blue), Fanconi
ATRA / arsenic	Differentiation syndrome → dexamethasone
5-FU / capecitabine	Coronary vasospasm (DPD deficiency)
Trastuzumab	Reversible cardiomyopathy

13. ICU Admission Decisions in Cancer

- Prognosis depends on **organ failures & performance status**, not cancer diagnosis alone
- **ICU trial:** full support 3–5 days then reassess
- Poor: > 3 organ failures after day 3, allo-HSCT with GVHD + MV
- Integrate **palliative care** early

14. High-Yield MCQ Pearls

- ✔ DCD requires no-touch period after asystole
- ✔ Brain-dead donor: vasopressin + desmopressin for DI
- ✔ Methylprednisolone 15 mg/kg in donors
- ✔ CNI toxicity: AKI, ↑K, PRES, TMA
- ✔ CMV peak 1–6 months post-transplant
- ✔ Denervated heart: atropine ineffective
- ✔ Neutropenic sepsis: anti-pseudomonal β-lactam ≤ 1 h
- ✔ Rasburicase contraindicated in G6PD deficiency
- ✔ CRS → tocilizumab; ICANS → steroids
- ✔ Leukostasis: avoid transfusion; leukapheresis
- ✔ ATRA → differentiation syndrome → dexamethasone
- ✔ Bleomycin → avoid high FiO_2